



EXTREMAL GRAPH THEORY

Erdős Problem 915

From Proof to Search

One-hour version

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The road

- 1 The question, and the one theorem it starts from
- 2 Four axes, sixteen variants
- 3 What falls to a hand proof
- 4 The directed turn, where the hand proof stops
- 5 Why a machine, and what it is allowed to claim
- 6 Discovery: tabu search
- 7 The landscape, the results, and what is left

Part 1

The question

Erdős Problem 915

How many edges can a graph on n vertices carry
if *no* pair of vertices has m edge-disjoint paths?

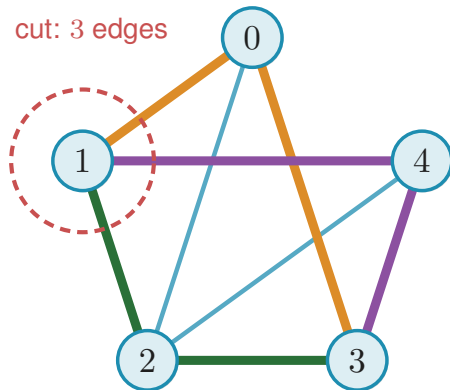
$$\ell_m(n) = \max \{ |E(G)| : |V(G)| = n, \lambda_G^{\max} \leq m - 1 \}$$

Mader, for all $n \geq 2$ and $m \geq 2$:

$$\ell_m(n) \leq \left\lfloor \frac{m(n-1)}{2} \right\rfloor$$

Posed by Bollobás and Erdős. The ceiling $\lambda_G^{\max} \leq m - 1$ says every pair can be severed by fewer than m edges, so Menger turns “many routes” into “costly cut”, and the whole thesis is built on that one exchange.

Menger, on a five-vertex example



Three edge-disjoint 1–3 routes meet the minimal three-edge cut around vertex 1, so $\lambda(1, 3) = 3$. A route count is a cut count.

The one tool: the Gomory–Hu tree

Every graph has a weighted spanning tree in which $\lambda_G(u, v)$ is the *lightest* edge on the tree path from u to v .

All $\binom{n}{2}$ local connectivities, stored in $n - 1$ numbers.

$\lambda_G^{\max}$ is simply the heaviest edge of that tree.

A double count on that tree proves Mader's bound, independently of Mader, and the same count returns for the multigraph value and for the hypergraph bound one edge size up. Digraphs have no Gomory–Hu tree, which is exactly where the hand proofs start to fail.

Part 2

Four axes, sixteen variants

The four axes

Arity

edge, on 2 vertices
hyperedge, on r

hypergraphs use Berge
routes

Multiplicity

simple
multi, with $\mu(u, v)$

the two model toggles are
independent, so four
models

Direction

undirected
directed

direction breaks the degree
symmetry that caps the
undirected answer

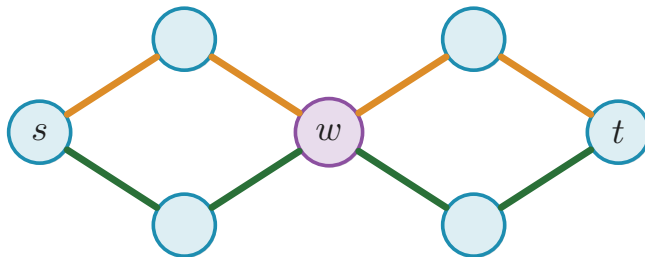
Separation

edge-disjoint, $\lambda^{\max}$
vertex-disjoint, $\kappa^{\max}$

Whitney: $\kappa \leq \lambda$, so the
vertex-feasible family is the
larger one

$4 \times 2 \times 2 = 16$ versions of one question

The separation axis



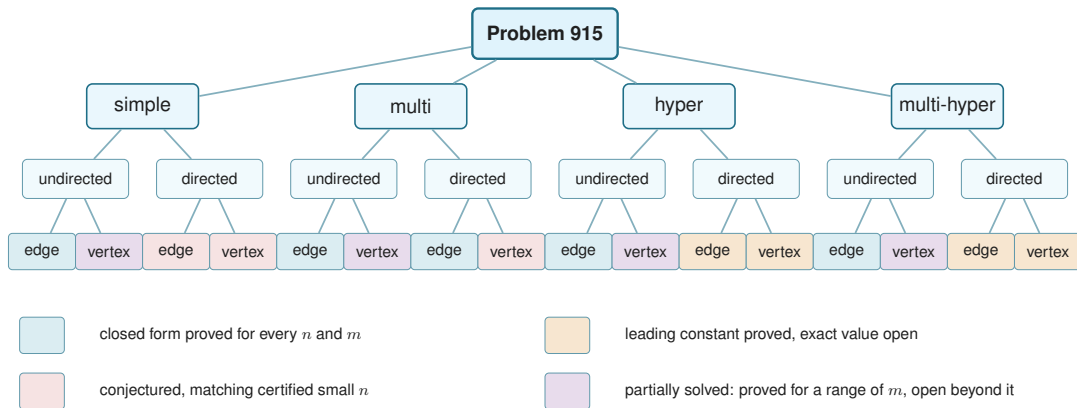
Two edge-disjoint s – t routes both pass through w , so $\lambda(s, t) = 2$ but $\kappa(s, t) = 1$. Parallel copies never raise κ , so the multigraph vertex rows are posed on the underlying simple graph.

Notation for the sixteen

Model	edge-disjoint routes	internally vertex-disjoint routes
Simple graph	$\ell_m(n)$	$k_m(n)$
Multigraph	$L_m(n)$	$K_m(n)$
r -uniform simple hypergraph	$\ell_m^{(r)}(n)$	$k_m^{(r)}(n)$
r -uniform multihypergraph	$L_m^{(r)}(n)$	$K_m^{(r)}(n)$
Simple digraph	$\ell_m^{\text{dir}}(n)$	$k_m^{\text{dir}}(n)$
Directed multigraph	$L_m^{\text{dir}}(n)$	$K_m^{\text{dir}}(n)$
Directed r -uniform simple hypergraph	$\ell_m^{\text{dir}(r)}(n)$	$k_m^{\text{dir}(r)}(n)$
Directed r -uniform multihypergraph	$L_m^{\text{dir}(r)}(n)$	$K_m^{\text{dir}(r)}(n)$

Lowercase for simple and uppercase for multi, with a dir superscript for one-way and an (r) superscript for hyperedges.

The sixteen leaves, and how much is known



Blue a closed form for every n and m , purple proved for a range of m and open beyond it, red a conjecture with a proved lower bound, amber a proved leading term.

Part 3

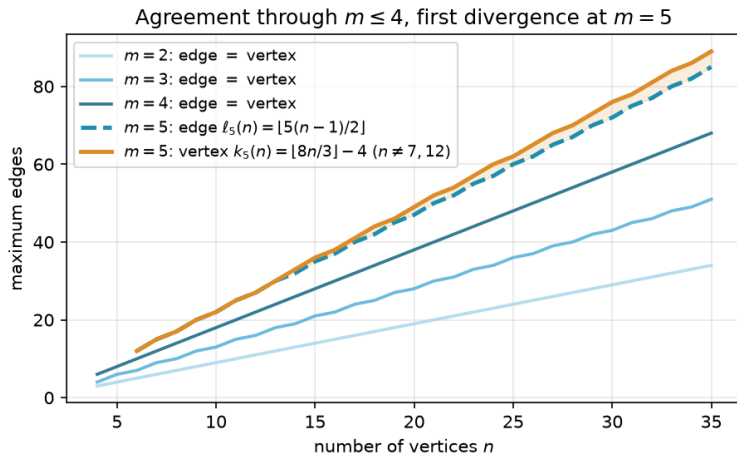
What falls to a hand proof

The short proofs

Simple graphs, edge-disjoint	$\ell_m(n) \leq \left\lfloor \frac{m(n-1)}{2} \right\rfloor$	(Mader, re-proved here, and attained for $n \geq m$)
Multigraphs, edge-disjoint	$L_m(n) = (m-1)(n-1)$	
Simple graphs, $m \leq 4$, $n \geq m$	$k_m(n) = \ell_m(n) = \left\lfloor \frac{m(n-1)}{2} \right\rfloor$	(Bárfai, Bollobás, Leonard)
Hypergraphs, edge-disjoint	$\leq \left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$	, attained in a stated range
Hypergraphs, vertex-disjoint at $m = 2$	$K_2^{(r)}(n) = \left\lfloor \frac{n-1}{r-1} \right\rfloor$	, and the star hypertree is simple

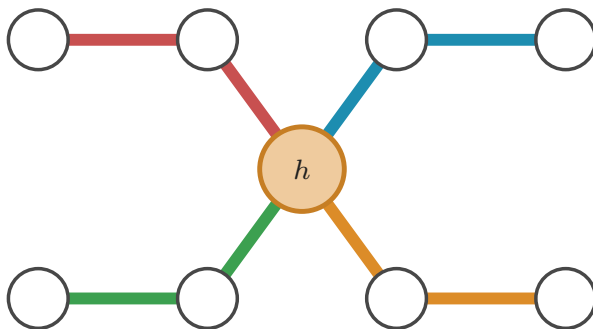
One argument serves the first, the second and the fourth: charge every edge to the Gomory–Hu tree.

The first divergence, at $m = 5$



Edge and vertex agree through $m = 4$. Sørensen and Thomassen give $k_5(n) = \lfloor 8n/3 \rfloor - 4$ for $n \geq 6$, $n \neq 7, 12$, which first passes the edge value at $n = 14$. Beyond $m = 5$ the vertex problem has stood open since 1974.

The star hypertree



$r = 3, n = 9$: four hyperedges meeting only at the hub, Berge connectivity one, attaining $\lfloor (n - 1)/(r - 1) \rfloor = 4$. One colour is one hyperedge, a line stopping at all r of its vertices and able to carry one route.

The hypergraph vertex problem

At $m = 2$, for every $r \geq 2$ and $n \geq 1$:

$$K_2^{(r)}(n) = \left\lfloor \frac{n-1}{r-1} \right\rfloor$$

vertex and edge agree at $m = 2$ for every uniformity, and repeated hyperedges never help

At $m = 3$, a universal upper bound for every r -uniform multihypergraph:

$$|E(\mathcal{H})| \leq \left\lfloor \frac{2(n-1)}{r-1} \right\rfloor$$

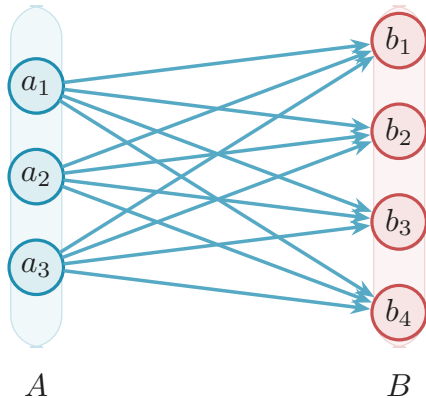
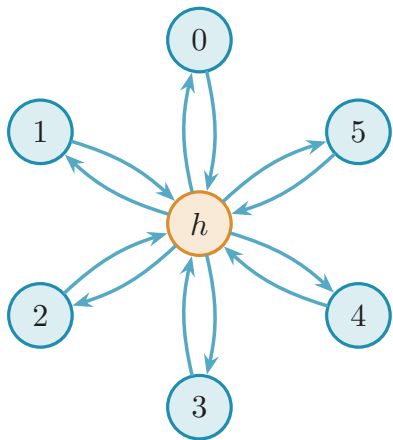
attained by a *simple* hypergraph for $r \geq 3$ once $2 \leq \binom{n-2}{r-2}$, so $k_3^{(r)}(n) = \ell_3^{(r)}(n)$ there, and by a multihypergraph whenever $(r-1) \mid (n-1)$

Both read Berge routes as ordinary paths in the incidence graph. The $m = 3$ case needs a new rank bound on that incidence graph, proved through Tutte's triconnected decomposition. At $m \geq 4$ the route stops for want of a 4-connectivity analogue, and the value is open.

Part 4

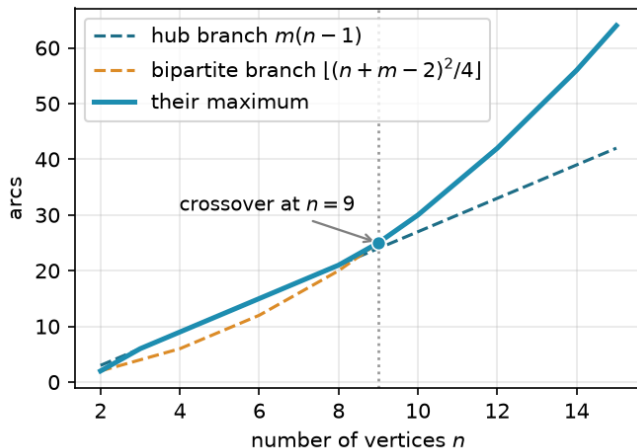
The directed turn

Two shapes tie at $n = 7$



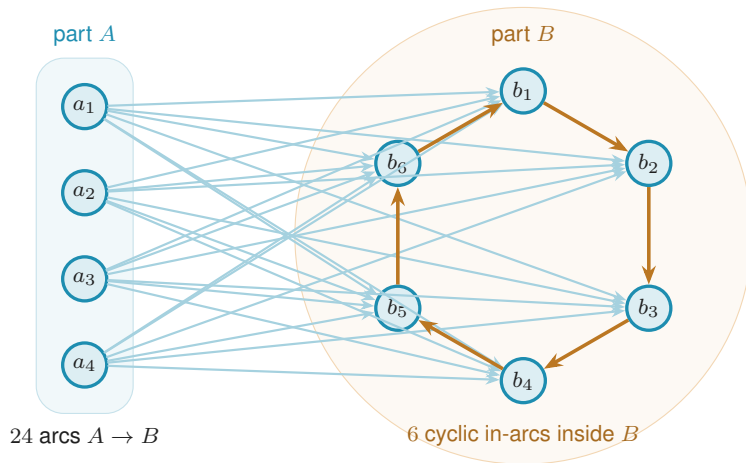
The bidirected star holds $2(n - 1) = 12$ arcs, the one-directional wall $B_{3,4}$ holds $\lfloor n^2/4 \rfloor = 12$, both at $\lambda^{\max} = 1$. Hence $\ell_2^{\text{dir}}(n) = \max(2(n - 1), \lfloor n^2/4 \rfloor)$, and $k_2^{\text{dir}}(n)$ is the same value.

Linear hub against quadratic wall



The hub spends its arcs on one centre and grows linearly. The augmented one-way wall grows quadratically and takes the lead near $n = 9$ when $m = 3$. No undirected graph can be this dense at a fixed connectivity ceiling.

The construction that carries the quadratic to every m



At $m = 3$, $n = 10$: every arc runs $A \rightarrow B$, and inside B each vertex takes $m - 2$ arcs from its cyclic predecessors. A pair a_i, b_j then has the direct arc and one detour, so $\lambda^{\max} = 2$, and $24 + 6 = 30$ beats the hub's 27.

The directed conjecture, and what is unconditional

Conjectured value, lower bound proved for all m , exhausted at $n \leq 6$ when $m = 3$:

$$\ell_m^{\text{dir}}(n) = \max\left(m(n-1), \left\lfloor \frac{(n+m-2)^2}{4} \right\rfloor\right)$$

Proved here with no hypothesis on the extremal shape, for both separations:

$$|A(D)| \leq \left\lfloor \frac{n^2}{4} \right\rfloor + 4(m-1)(n-1)$$

and therefore, for each fixed $m \geq 3$,

$$\ell_m^{\text{dir}}(n), k_m^{\text{dir}}(n) = \frac{n^2}{4} + \Theta_m(n)$$

The engine is a two-step route count: a vertex cannot have many out-neighbours that reach one another, or the short detours breach the ceiling. Leading constant and order of the second term are both theorems. What is left is one coefficient, $(m-2)/2$ conjectured against $4(m-1)$ proved.

Shape as well as size

In size. The extremal value lies within $O_m(n)$ of the wall's count, so what is open is a constant inside the second-order term rather than the phenomenon.

In shape. Every feasible digraph becomes a subgraph of a one-directional bipartite graph after deleting $O(\sqrt{m} n^{3/2})$ arcs. A weaker error term, because it comes from the count that does not induct on a low-degree vertex.

What a full upper bound still needs: find the partition, rule out the arcs running back, and control the internal structure of the larger part B . One returning arc alone breaks the count, since it lets a route leave B through A and exceed the internal budget of $m - 2$. Ruling that arc out for every extremal example is the missing step.

One-way multigraphs, solved outright

$$L_m^{\text{dir}}(n) = (m - 1) M(n), \quad M(n) = \max\left(2(n - 1), \left\lfloor \frac{n^2}{4} \right\rfloor\right)$$

for every $n \geq 2$ and every $m \geq 2$

The proof deletes a *subgraph* at a time rather than a vertex. Inside any feasible multigraph there is a sparse skeleton that alone preserves which vertices reach which, and removing one copy of each of its arcs costs every pair at least one route. Peeling off $m - 1$ such skeletons accounts for every arc, and the parity obstruction that stopped the vertex-deletion induction never arises.

No parity split and no auxiliary conjecture. Equality also classifies the quadratic extremisers once $n \geq \max(8, 2m)$, while the linear-branch extremisers are exhibited rather than classified. The finite $n = 7$ fact this problem once needed a solver for now follows in three lines.

One-way hyperedge networks

For every forward directed r -uniform multihypergraph with $\kappa_{\mathcal{H}}^{\max} \leq m - 1$:

$$|E(\mathcal{H})| \leq \frac{m-1}{r-1} \left\lfloor \frac{n^2}{4} \right\rfloor + 4(m-1)^2(n-1)$$

so for fixed r and m , under *both* separations,

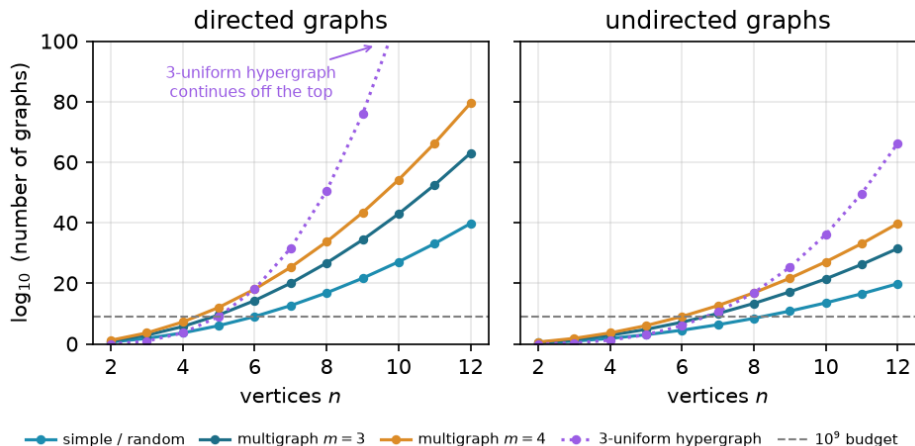
$$(1 + o(1)) \frac{(m-1) n^2}{4(r-1)}$$

The proof forms the one-step shadow digraph, whose arc $u \rightarrow v$ records that some hyperedge has tail u and head v . A matching argument transfers the route ceiling to the shadow's two-step budget, and a tail-head incidence count converts shadow arcs back to hyperedges. The bipartite construction already attains that leading term, so it is asymptotically optimal. Before this, the bounds were a factor of four apart.

Part 5

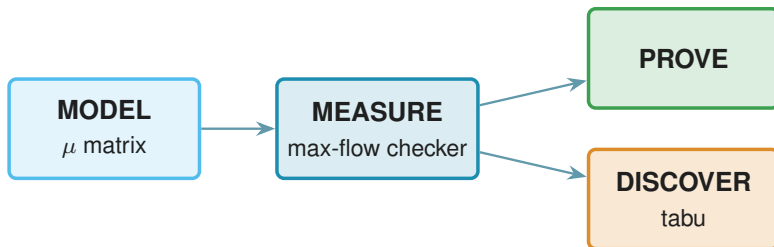
Why a machine

Enumeration detonates



A simple graph offers 2^E candidates, a multigraph m^E , and a digraph decides both directions of a pair independently. Every curve passes a generous 10^9 budget while n is in single digits, and the directed cases go first.

Three jobs, kept apart



One model, one exact measurement, two very different uses of it. The badge colours are the thesis's, and every code card in Chapter 2 carries the badge of the box it belongs to.

What each job is allowed to claim

MEASURE

One integer maximum flow, on one multiplicity matrix or one hyperedge list, for all sixteen variants.

What it reports about a given graph is a fact.

PROVE

Exhaust every graph of a fixed size, pruned, or seeded by geng.

A finished sweep is an upper bound at that size. It is a finite solver check, not a reusable certificate.

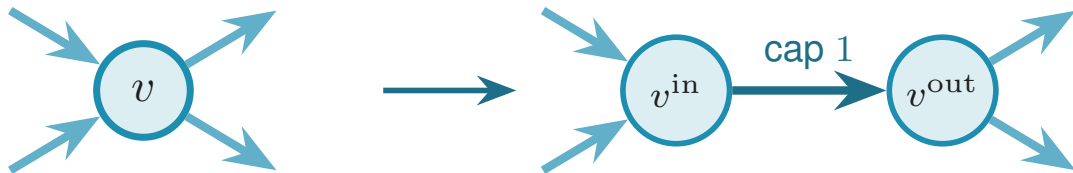
DISCOVER

Climb with a tabu list, from the empty graph, restarted from many seeds.

Only ever exhibits a graph. Lower bounds only. Never a proof.

The search proposes a lower bound, a proof or a finished exhaustion disposes the upper bound,
and only when the two meet is a value called settled. No theorem here rests on an unchecked solver result.

One flow engine for sixteen variants



Every vertex-disjoint question becomes an edge-disjoint one: split v into v^{in} and v^{out} joined by a unit-capacity arc, so two routes through v compete for that unit. Undirected edges become opposite arc pairs, and each hyperedge becomes a gate whose single internal arc carries its multiplicity.

Proving a fixed size, three ways

n	$\ell_2^{\text{dir}}(n)$	blind (s)	pruned (s)	generated (s)
3	4	0.05	0.003	0.009
4	6	6.3	0.008	0.008
5	8	—	0.22	0.039
6	10	—	14.5	0.73
7	12	—	1156.6	17.0

Wall-clock time to prove $\ell_2^{\text{dir}}(n)$ on one core. The blind sweep walks every matrix assignment, and it is dropped past $n = 4$, where $2^{n(n-1)}$ assignments already cost 6.3 s and each further vertex multiplies that by 2^{2n} . The pruned search stops a branch the moment a pair exceeds the ceiling, since adding arcs never lowers connectivity. The generated pipeline calls `geng` from `nauty` for one representative per isomorphism class and decorates its edges with arc multiplicities, inheriting the isomorphism reduction. From $n = 5$ the gap widens every vertex: 68 times faster at $n = 7$. All three return the same proved value wherever they finish.

Proving every m at once

Divide every multiplicity of a feasible directed multigraph by $m - 1$.

weights $w(u, v) \in [0, 1]$, pairwise maximum flow ≤ 1 ,
total weight = arc count / $(m - 1)$

One m -free optimisation therefore bounds every threshold at once, for a fixed n . The general theorem $L_m^{\text{dir}}(n) = (m - 1)M(n)$ has since replaced this route in the solver, but the older formulation is kept as an independent finite check, and the encoding is exact only up to $n = 6$: at $n = 7$ the certifier ran out of time without an answer.

Part 6

Discovery

One number per graph

$$\Phi(G) = -|E(G)| + \Lambda \cdot \max(0, c^{\max}(G) - (m - 1))$$

rewards edges, arcs or hyperedges, penalises any excess connectivity, with $c^{\max}$ standing for $\lambda^{\max}$ or $\kappa^{\max}$, whichever the run measures

Tabu search. Start from the empty graph and toggle one adjacency at a time. Among the moves not forbidden, always take the one that lowers Φ most, *even when every candidate raises it*. The pair just changed goes on a tabu list for a fixed number of steps, so the walk cannot immediately undo itself and cycle back. A move that would beat the best graph seen overrides the tabu flag.

Why the walk is allowed above the ceiling: a feasible graph can be locally maximal and still far below the densest one. Improving it needs an add and then a different removal, a swap whose first half is infeasible. Going above the bound is never the goal. It is the shortcut on the way back down to it. The checker is also skipped whenever monotonicity already fixes the answer, so a flow is computed only for an addition made *at* the ceiling.

Tabu against annealing

Variant		target	tabu	annealing
directed multigraph	$n = 7, m = 3$	24	24	20
simple digraph	$n = 7, m = 3$	18	18	16

Simulated annealing runs on the same energy and the same neighbourhood, accepting a worsening move by the Metropolis rule and cooling as it goes. At equal wall-clock budget it is the weaker engine here: tabu takes fewer, better-aimed steps and assembles the structured extremiser the cooled walk cannot reach. Every discovery value the thesis reports therefore comes from tabu search, and annealing is kept only as an independent cross-check in the self-test, where agreement between two unrelated walks is worth more than the speed of either.

Validation by rediscovery

Construction	Variant	n	m	Search	Theory
spanning tree (simple undirected)	$\ell_m(n)$	6	2	5	5
multiplicity- $(m-1)$ star (multi undirected)	$L_m(n)$	6	3	10	10
bidirected star (simple directed)	$\ell_m^{\text{dir}}(n)$	4	2	6	6
bidirected star (simple directed)	$\ell_m^{\text{dir}}(n)$	6	2	10	10
hub-bipartite tie (simple directed)	$\ell_m^{\text{dir}}(n)$	7	2	12	12
double star (multi directed)	$L_m^{\text{dir}}(n)$	4	3	12	12
double star (multi directed)	$L_m^{\text{dir}}(n)$	5	3	16	16

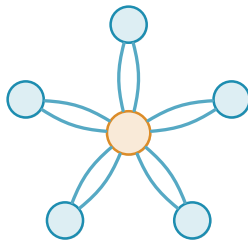
Before the search is pointed at the open cases it is run on the settled ones. From scratch, with a handful of restarts, and told only to pack in edges without breaching the ceiling, it recovers the named constructions and matches the established value in every tested case.

The shapes it comes back with



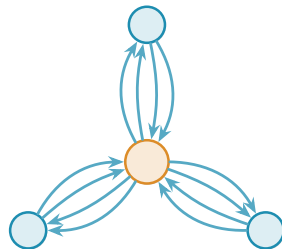
spanning tree

$$n = 6, \ell_2(n) = 5$$



multigraph star

$$n = 6, m = 3, L_3(n) = 10$$



double star

$$n = 4, m = 3, L_3^{\text{dir}}(n) = 12$$

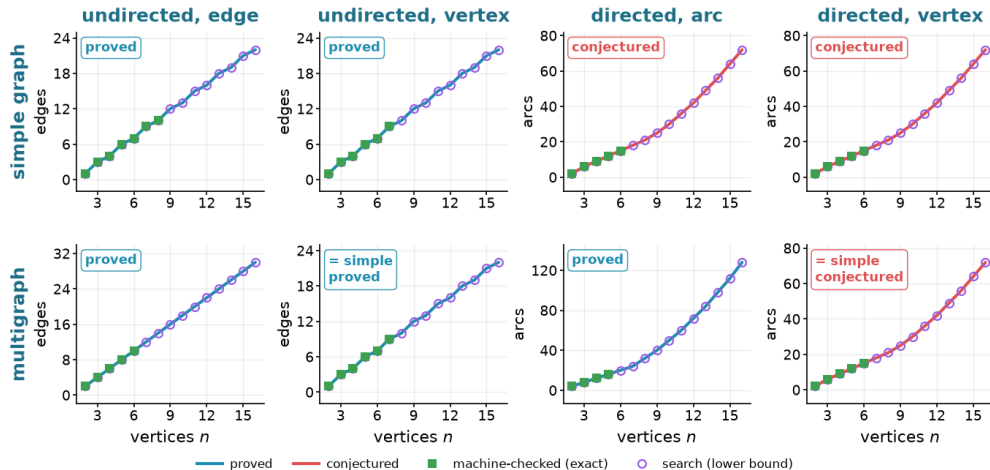
Three of the constructions the table names, drawn at the size it reports. The fourth is the bidirected star that ties the one-directional wall at $n = 7$.

Part 7

The landscape, and what is left

The eight graph-model variants at $m = 3$

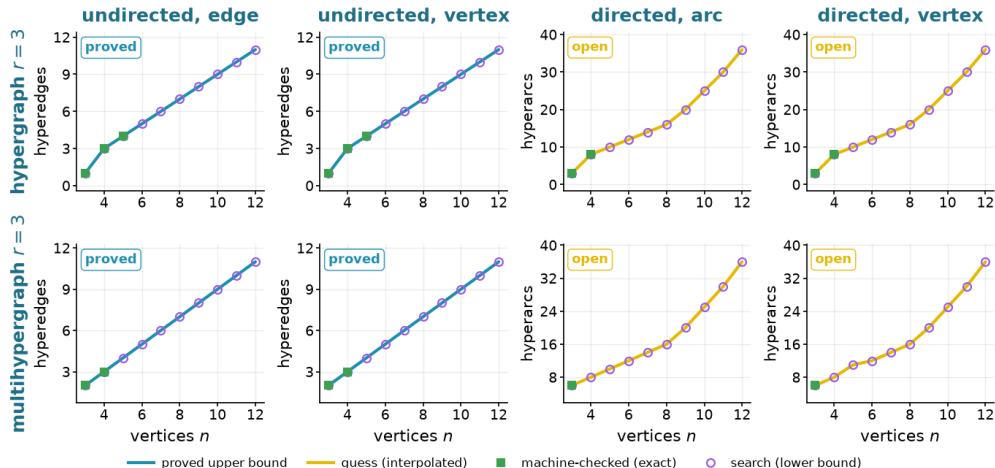
Erdős 915 across the sixteen variants, $m = 3$ · simple and multigraph models



Blue a theorem, red a conjecture, yellow an interpolation. Filled squares are exact, open circles are witnesses.

The eight hypergraph-model variants at $m = 3$

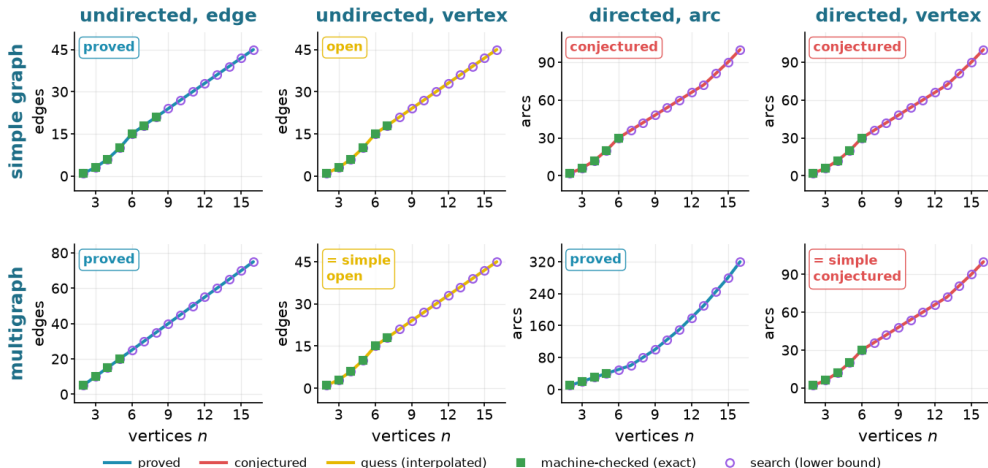
Erdős 915 across the sixteen variants, $m = 3$ · hypergraph and multihypergraph models



The blue curve here is a proved upper *bound*, attained only in a stated range, so a marker below it is a gap and not an error.

The same graph models at $m = 6$

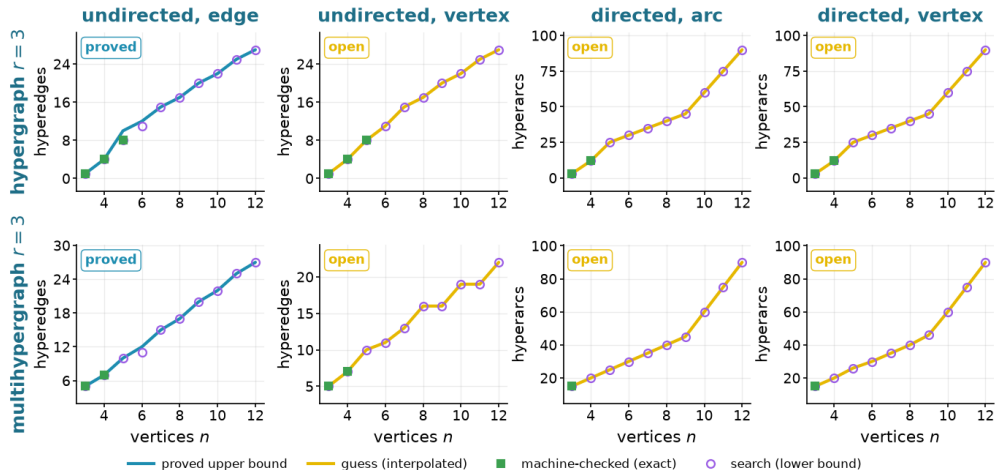
Erdős 915 across the sixteen variants, $m = 6$ · simple and multigraph models



Raising the threshold widens every feasible band, and the exact squares thin out as enumeration falls behind the constructions.

The same hypergraph models at $m = 6$

Erdős 915 across the sixteen variants, $m = 6$ · hypergraph and multihypergraph models



Here the bound runs visibly above what is reached: at $n = 6$ the curve reads 12 where the true maximum, by full enumeration, is 11.

Where each variant stands

Model	Edge or arc-disjoint	Vertex-disjoint
Undirected		
simple graph	$\left\lfloor \frac{m(n-1)}{2} \right\rfloor$ PROVED Mader	the edge value for $m \leq 4$, and $\lfloor 8n/3 \rfloor - 4$ at $m = 5$ RANGE open $m \geq 6$
multigraph	$(m-1)(n-1)$ PROVED	the simple value RANGE parallel copies do not raise κ
r -hypergraph	$\leq \left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$, equality in the stated range PROVED	$\left\lfloor \frac{n-1}{r-1} \right\rfloor$ at $m = 2$, $\leq \left\lfloor \frac{2(n-1)}{r-1} \right\rfloor$ at $m = 3$ RANGE open $m \geq 4$
r -multi-hyper	the same bound, equality when $(r-1) \mid (n-1)$ PROVED	the same bounds, repeats closing cases at $m = 3$ RANGE
Directed		
simple digraph	$M(n)$ at $m = 2$; $n^2/4 + \Theta_m(n)$ for $m \geq 3$ PROVED $m = 2$ CONJ $m \geq 3$	$M(n)$ at $m = 2$; $n^2/4 + \Theta_m(n)$ for $m \geq 3$ PROVED $m = 2$ CONJ $m \geq 3$
multigraph	$(m-1)M(n)$ PROVED every n and m	the simple digraph value CONJ $m \geq 3$
r -hypergraph	$(1 + o(1)) \frac{(m-1)n^2}{4(r-1)}$ LEADING TERM	the same leading term LEADING TERM
r -multi-hyper	the same leading term, repeats allowed LEADING TERM	the same leading term LEADING TERM

$M(n) = \max(2(n-1), \lfloor n^2/4 \rfloor)$, and the simple-graph formulas assume $n \geq m$. **PROVED** closed form for every n and m . **RANGE** proved for a range of m , open beyond it. **CONJ** conjectured formula, lower bound proved and the small cases exhausted. **LEADING TERM** the n^2 term is a theorem, the exact value is not.

What is new here

- 1 One-way multigraphs, solved outright:** $L_m^{\text{dir}}(n) = (m - 1) \max(2(n - 1), \lfloor n^2/4 \rfloor)$ every n and every m , by peeling off one sparse reachability-preserving subgraph per unit of $m - 1$, with the quadratic-branch extremisers classified for $n \geq \max(8, 2m)$.
- 2 One-way simple graphs: an unconditional $n^2/4 + \Theta_m(n)$, arc and vertex alike**
no assumption about the extremal shape, plus an $O(\sqrt{m} n^{3/2})$ structural distance from a one-way bipartite graph. The vertex statement is not a corollary of the arc one.
- 3 The hypergraph vertex problem: exact at $m = 2$, and sharp at $m = 3$ in its construction range**
a universal $m = 3$ upper bound from a new incidence-graph rank lemma, with equality for $r \geq 3$ once $2 \leq \binom{n-2}{r-2}$.
- 4 Multigraph vertex routes with parallel copies counted as distinct**
exact at $m = 2$, counterexamples to the thickened-tree guess for every $m \geq 5$, matching order $\Theta(m^2 n)$, and an exact block-knapsack formulation.
- 5 One-way hyperedge networks: the leading term settled**
 $(1 + o(1))(m - 1)n^2/(4(r - 1))$ for fixed r, m , under both separations and every orientation convention, so the bipartite construction is asymptotically optimal.

What is left

Problem	Known here	Remaining target
Directed simple arc, $m \geq 3$	$\ell_m^{\text{dir}}(n) = n^2/4 + \Theta_m(n)$	Prove the decomposition, or otherwise determine the linear coefficient.
Directed simple vertex, $m \geq 3$	$k_m^{\text{dir}}(n) = n^2/4 + \Theta_m(n)$	Decide whether it equals the arc value, given that an arc upper bound does not transfer.
Directed multigraph classification	Exact value for all n, m and quadratic-branch uniqueness for $n \geq \max(8, 2m)$	Classify the range $n < 2m$ and the remaining linear-branch extremisers.
Undirected simple vertex, $m \geq 6$	Block formulation and $m/2 < c_m \leq 2(m-1)$, where $c_m = \lim_n k_m(n)/(n-1)$	Determine c_m , whose likely next structure is triconnected rather than merely blockwise.
Multigraph vertex, parallel copies distinct	Block-knapsack formulation and order $\Theta(m^2 n)$, per-vertex constant between $3 - 2\sqrt{2}$ and 2	Determine the asymptotic constant and the optimal 2-connected blocks.
Hypergraph vertex, $m \geq 4$	Exact at $m = 2$ and, in the stated range, $m = 3$	Locate the first failure for $r \geq 3$ and replace the rank-two decomposition used at $m = 3$.
Directed hypergraphs	For fixed r, m , leading term $(m-1)n^2/(4(r-1))$, all orientations, both separations	Determine finite exact values and whether forward and general agree once $n > r$.

The map is filled in where the arguments reached, and the blanks are marked where they did not.

Backup

Further material for the question round

The sixteen variants as numbers

Model	Variant	$n = 2$	$n = 3$	$n = 4$	$n = 5$	$n = 6$	$n = 7$	$n = 8$
$m = 3$								
simple graph	undirected, edge $(\ell_m(n))$, proved	1	3	4	6	7	9	10
	undirected, vertex $(k_m(n))$, proved	1	3	4	6	7	9	10
	directed, arc $(e_m^{dir}(n))$, conjectured	2	6	9	12	15	≥ 18	≥ 21
	directed, vertex $(k_m^{dir}(n))$, conjectured	2	6	9	12	15	≥ 18	≥ 21
multigraph	undirected, edge $(L_m(n))$, proved	2	4	6	8	10	12	14
	undirected, vertex $(k_m(n))$, = simple, proved	1	3	4	6	7	9	10
	directed, arc $(L_m^{dir}(n))$, proved	4	8	12	16	20	24	32
	directed, vertex $(k_m^{dir}(n))$, = simple, conjectured	2	6	9	12	15	≥ 18	≥ 21
hypergraph $r = 3$	undirected, edge, proved	1	3	4	≤ 5	≤ 6	≤ 7	
	undirected, vertex, proved	1	3	4	≤ 5	≤ 6	≤ 7	
	directed, arc, open	3	8	≥ 10	≥ 12	≥ 14	≥ 16	
	directed, vertex, open	3	8	≥ 10	≥ 12	≥ 14	≥ 16	
multihypergraph $r = 3$	undirected, edge, proved	2	3	≤ 4	≤ 5	≤ 6	≤ 7	
	undirected, vertex, proved	2	3	≤ 4	≤ 5	≤ 6	≤ 7	
	directed, arc, open	6	≥ 8	≥ 10	≥ 12	≥ 14	≥ 16	
	directed, vertex, open	6	≥ 8	≥ 11	≥ 12	≥ 14	≥ 16	
$m = 6$								
simple graph	undirected, edge $(\ell_m(n))$, proved	1	3	6	10	15	18	21
	undirected, vertex $(k_m(n))$, open	1	3	6	10	15	18	≥ 21
	directed, arc $(e_m^{dir}(n))$, conjectured	2	6	12	20	30	≥ 36	≥ 42
	directed, vertex $(k_m^{dir}(n))$, conjectured	2	6	12	20	30	≥ 36	≥ 42
multigraph	undirected, edge $(L_m(n))$, proved	5	10	15	20	25	30	35
	undirected, vertex $(k_m(n))$, = simple, open	1	3	6	10	15	18	≥ 21
	directed, arc $(L_m^{dir}(n))$, proved	10	20	30	40	50	60	80
	directed, vertex $(k_m^{dir}(n))$, = simple, conjectured	2	6	12	20	30	≥ 36	≥ 42
hypergraph $r = 3$	undirected, edge, proved	1	4	8	≤ 12	≤ 15	≤ 17	
	undirected, vertex, open	1	4	8	≥ 11	≥ 15	≥ 17	
	directed, arc, open	3	12	≥ 25	≥ 30	≥ 35	≥ 40	
	directed, vertex, open	3	12	≥ 25	≥ 30	≥ 35	≥ 40	
multihypergraph $r = 3$	undirected, edge, proved	5	7	≤ 10	≤ 12	≤ 15	≤ 17	
	undirected, vertex, open	5	7	≥ 10	≥ 11	≥ 13	≥ 16	
	directed, arc, open	15	≥ 20	≥ 25	≥ 30	≥ 35	≥ 40	
	directed, vertex, open	15	≥ 20	≥ 26	≥ 30	≥ 35	≥ 40	

Blue proved, red conjectured and gold open, and a bold green entry is machine-checked exact regardless of the row's colour. A $\leq$ prefix bounds a value not known to be attained, a $\geq$ prefix is a verified construction rather than a proved optimum, and a blank cell means the model does not exist at that n .

Orientations of a directed hyperedge

The tables use the **forward** convention: one tail and $r - 1$ heads. Reversing every hyperedge makes the one-head **backward** convention equivalent to it. And the same leading term extends to *every* other way of splitting a hyperedge into tail and head sets, under both separations.

The orientation therefore adds no asymptotic row.

The conventions may still differ at few vertices, where $n \approx r$. Whether they agree exactly once $n > r$ is open, so Problem 915 stays a sixteen-variant question.

Limits of the program

- The search finds examples. It never proves an upper bound.
- The certifier proves an upper bound at one fixed size, as a finite solver check rather than a replayable certificate.
- The directed multigraph encoding is exact only to $n = 6$. At $n = 7$ the certifier ran out of time, and the value comes from the hand proof instead.
- The $n = 7$ generation classification is verified only for extremisers of maximum total degree at most 8.

Three of the remaining gaps cannot be closed by computation at all: the decomposition behind the directed conjecture, the directed vertex problem at $m \geq 3$, and the undirected vertex problem at $m \geq 6$. Every computational claim can be rerun from `program/erdos915_unified.py`, and the important values are checked by independent implementations.